

Writeup

Rationale for design process:

After some exploration of the CMIP6 data, we realized that the tos dataset storing the Sea Surface Temperature for monthly ocean data could be represented using different ocean regions. Since the historical data captures experiments run between 1850 up until the end of 2015, we wanted to intentionally split up these time periods to show the differences in the stats we could capture over each time frame. Considering the fact that modern recordkeeping of ocean temperature data didn't begin until the 1950s, we decided to use 1950-1989 as our baseline. Based on ocean heat content changes recorded by the NOAA, we noticed that the steepest temperature increase was captured from the 1990s to present, and presented the change from our baseline using these two periods in our interactive visualization. Using the coordinates, we grouped the temperature data by ocean region as recognized by the national oceanography centre.

We decided upon the geoNaturalEarth1 projection from the d3 library as it allows for limited distortion around the Arctic and Southern Ocean near the poles and is the most visually accurate when dividing along the latitude and longitude lines. The text at the top of the visualization provides a basic explanation of what the visualization represents, and how viewers can use it. Our primary title isn't too specific, as our visualization represents two different analyses, and the viewers can switch between the two. Thus, the primary title provides a more holistic view of the graph. The smaller, centered text right above each visual provides more specific information: ocean temperature by region, and change in ocean temperature over a specific time span. The first part of the visual, representing ocean temperatures by region, implements a categorical color scale as explained in the legend. Each color corresponds to the average temperature for each region of the ocean and distinctly categorizes each region, which is why we implemented a categorical color scale. To display more specific stats, we use d3 to allow for the viewer to hover over the region to view the minimum, average, and maximum temperature over this time period. In addition to the ocean temperatures, viewers can use the show change button to visualize the regions that have had a positive or negative change in average ocean temperature amongst historical and present day temperatures. We used a diverging color scale for this part of the visualization as it best encompasses the idea of whether the ocean is cooling down or warming up. Immediately, viewers can see how most of the map is some shade of red, representing how many of the oceans are indeed warming up. Similarly to the first part, in order to get more specific statistics about each region, viewers can use the toggle function to get the historical mean temperature, present mean temperature, and difference.

Overview of development process:

We split the development process into two iterative parts: EDA and interactive visualizations. While two of us started on analysis with EDA, one of us began on the interactive section for the working components of drawing the map using the d3 json file for countries and map projections. Together, we analyzed which variable, experiment, table, and source ids were compatible after deciding on Sea Surface data and identified the MIROC6 model as a working model for that dataset. It took about two hours to fully organize the data, interpret the variables, and locate the working model. We used time series plots to map the change in monthly sea surface temperature over the entire time span of the data and noticed that there was a gradual

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increase in surface temperatures over the years and a steep increase in sea level rise over the same years. We focused on analysis over the hottest ocean regions using a spatial mapping corresponding to the temperatures, motivating our idea to map temperatures over different regions while capturing the change in temperature that we had seen in the time series plots. After running simple statistics on each ocean region, we wanted to show how this change could be reflected in more recent data to baseline average temperatures. Using these statistics, the member starting the d3 process was able to use a hover tooltip to show these statistics, encode color as the temperature, and recommend additional EDA steps to show the changes in these temperatures. This allowed for our group to revisit EDA together and collectively add features, including the toggle button to reflect the delta, as well as the new color encoding. The interactive visualization process took about two days in total; the first day was used to correctly align the regions and add the tooltip, and the second day was used to add the button and new legend to show the change.

Citations:

NASA. (2025, September 26). *Ocean warming - earth indicator - NASA science*. NASA.
<https://science.nasa.gov/earth/explore/earth-indicators/ocean-warming/>